

Junghun Hwang (황정헌)

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Education

Konkuk University

Mar 2022 - Expected Feb 2027

B.S. in Smart Mobility Engineering | Seoul, Korea

Completed coursework: Linear Algebra, Autonomous System Platform, Artificial Intelligence Applications, Data Structures and Algorithms

In progress: AI Mobility Systems

Research Interests

Autonomous-driving perception and decision-making, with a focus on learning-based and end-to-end driving. Interested in how occlusion and input delay affect driving behavior, and in evaluating these effects in closed-loop experiments.

Research & Project Experience

2026 9th Kookmin University Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026

Team Lead | Team of 5

- Built a ROS 2 imitation-learning pipeline for steering, covering data collection, training, and TorchScript inference; implemented switching between the learned policy and rule-based mission logic.
- Prepared and reviewed Roboflow datasets and trained a unified YOLO detector for traffic lights, vehicles, obstacles, and cones.
- Integrated mission-management and driving modules and tuned the competition vehicle. Persistent steering oscillation in the learned policy led the team to use rule-based control for the final competition.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.

Team results: Preliminary 2nd Place | Finals 13th Place (course completed) | Parking Special Award

2026 Student AI · SW Mobility Competition

Mar 2026 - Ongoing

Team of 27 | Decision-Making Subteam (5)

- Developed ROS 2 cone-boundary path generation, red-cone-triggered stop commands, and stateful fail-safe logic; implemented Stanley tracking with steering limits and motion preview.
- Tested the AEB pipeline in closed-loop Gazebo simulation and inspected path, steering, and stop states in Foxglove.
- Completed real-vehicle waypoint runs on both preliminary and final courses at 10 m/s (36 km/h) during venue practice on Sep 19-20, 2026.
- In a 50 km/h AEB trial, confirmed a stop command at the red cones; a brake hardware fault prevented stopping within the required zone.

Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing

Capstone Design | Requirements & System Design; Implementation Planned

- Defined a daytime, low-speed pedestrian-occlusion ODD for a 1/5-scale HENES platform and a matched comparison of camera-only and V2I-assisted E2E driving.
- Specified a road-plane homography pipeline and direct transmission of pedestrian position, confidence, and timestamps for lightweight fusion into waypoint and target-speed outputs.
- Formulated a detection-distance budget from latency, braking distance, and a safety margin. Planned measurements of stagewise latency, earlier pedestrian detection and deceleration, minimum TTC, and collisions.

ROS 2 UAV Waypoint Mission on PX4 SITL/Gazebo

May 2026 - Jun 2026

Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 mission package with CSV waypoints, TF-based arrival checks, and coordinate-mode handling; integrated it with an existing PX4 offboard controller via PoseStamped commands.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

ROS 2 / CARLA Decision and Control

Mar 2026 - Jun 2026

Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated unstable simulated tracking through steering direction, lookahead, smoothing, and command limits. The root cause remained unresolved; stable autonomous driving and a real-vehicle run were not achieved.

Awards & Competitions**2026 Kookmin AI-Edge Challenge**

Parking Special Award (team)

2022 Drone Clash, General Division

3rd Place

Technical Skills**Languages:** Python, C++**Autonomous Systems:** ROS 2, CARLA, Gazebo, PX4 SITL**AI / Computer Vision:** PyTorch, YOLO, OpenCV, Roboflow**Tools / Environment:** Ubuntu, Git/GitHub, Foxglove, RViz2, rosbag2, QGroundControl